BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 060 · Lesson 060 · Refining an Abstraction and Testing Assumptions

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does abstraction remove?

Q2. What is a model assumption?

Q3. What is a boundary test?

Q4. 031

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

refinement — Improving model for new requirements.

boundary — Important threshold or edge.

validity — Whether model works for its intended use.

assumption — Rule taken as given.

New learning

Understand — Compare the simplification against new requirements. If the model omits a relevant factor it can produce plausible but wrong outputs.

Model — Refine the abstraction by adding only the newly necessary variable, rather than reconstructing every real-world detail.

Evaluate — Test normal, boundary and exceptional cases: 5 km versus 6 km, return and single, and invalid negative distance.

Worked example

A simplified cinema ticket model uses only age but now weekend prices differ. What change is needed?

1. Explain — Weekend becomes a relevant distinction because it can change the output.

We Do · Practise with guidance

A parking price model charges £3/hour but introduces £12 maximum. Improve algorithm.

```
fee=MIN(3×hours,12), with validation that hours≥0.
```

Explain

Cap is new relevant rule. Example 5 hours costs £12, not £15.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Ticket v1 charges £2 at ≤5km, £4 otherwise. Give charge at 5km and 6km and why these are useful tests. [6]

Question 2

Return journeys cost 1.8 times single. For 8km journey calculate return fare and show variables. [6]

Question 3

A school model predicts homework concerns from just missed work counts; discuss which relevant conditions to add and how to test impact. [6]

Exit Ticket · Answer before checking

A model that worked yesterday never needs changing when requirements change.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Irrelevant details.

Q2. Simplifying rule treated as true for a task.

Q3. An input at/around an important threshold.

Q4. Rising-edge D-type has Q=1 and D=0 before edge. Give Q just after edge.

You Do pages · Independent questions

Q1. 5km £2 [2],6km £4 [2]; tests either side of threshold [2].

Q2. Single fare 8>5→£4 [2], return multiplier1.8 [2], total £7.20 [2].

Q3. Consider absence/accessibility/extension [2], avoid inappropriate irrelevant personal factors [1], human review [1], test representative boundary cases [2].

Exit Ticket · Model answer

False. New relevant factors can invalidate assumptions.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: